Rayoscan AI Diagnostics Assignment Report

Prithvi Raj Kaushik Metpally

MTech Medical Device Innovation, IIT Hyderabad

1. Introduction

This project was carried out as part of the AI Intern selection process for Rayoscan AI Diagnostics. The objective was to develop an object detection model capable of identifying and classifying tooth numbers from annotated dental images.

To achieve this, a YOLO (You Only Look Once) based deep learning model was trained on the provided dataset consisting of approximately 500 dental images with bounding box annotations.

2. Problem Statement

The goal of this assignment is to:

Detect teeth in dental images

Classify each detected tooth into its respective tooth number category

Build a complete pipeline including training, validation, and inference

This is a multi-class object detection problem involving 32 distinct classes, each corresponding to a specific tooth number.

3. Dataset Description

The dataset contains:

- ~500 annotated dental images
- Bounding box annotations in YOLO format
- 32 distinct tooth classes
- Dataset Structure
- The dataset was organized into:
- Training set: 80%

- Validation set: 10%
- Test set: 10%

4. Data Preprocessing and Validation

Before training, the dataset was thoroughly validated to ensure compatibility with YOLO.

Steps performed:

- Verified image-label consistency
- Ensured all labels follow YOLO format (5 values per line)
- Checked for invalid or corrupted annotations

Key Observation

During validation, 3 annotation files were found to be corrupted, containing a mix of:

- Bounding box annotations
- Polygon (segmentation-style) annotations

These files were incompatible with YOLO detection training.

Action Taken

- Removed the 3 corrupted image-label pairs from the dataset
- Ensured the remaining dataset was clean and consistent
- This preprocessing step was critical to ensure successful model training.

5. Model Selection

For this task, the YOLO11n model from the Ultralytics framework was selected because:

- Efficient and lightweight model
- Suitable for small datasets
- Fast training and inference
- Strong baseline performance for object detection tasks

6. Training Configuration

The model was trained with the following configuration:

- Model: YOLO11n
- Image size: 640 × 640
- Epochs: 30
- Batch size: 16
- Pretrained weights: Yes
- Early stopping patience: 10
- Platform: Google Colab
- Hardware: CPU (Intel Xeon)
- Due to computational constraints, training was performed on CPU.

7. Results

Final model performance after 30 epochs:

- Precision: 0.417
- Recall: 0.772
- mAP@50: 0.514
- mAP@50–95: 0.354

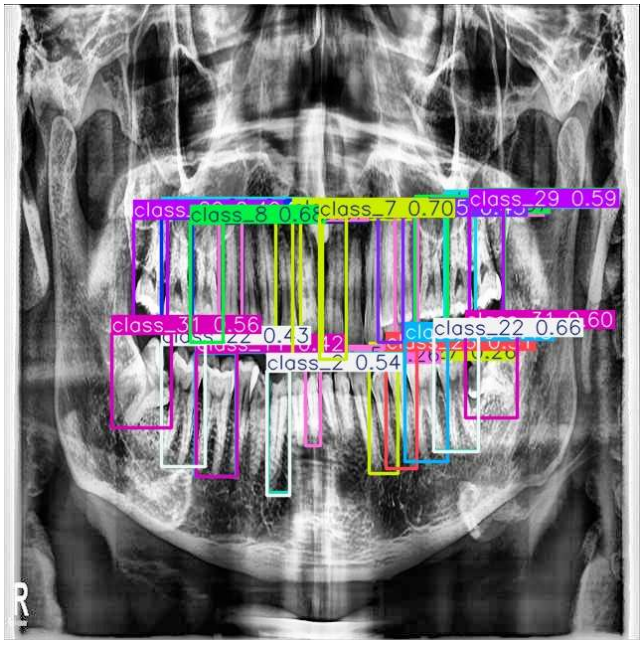
Interpretation

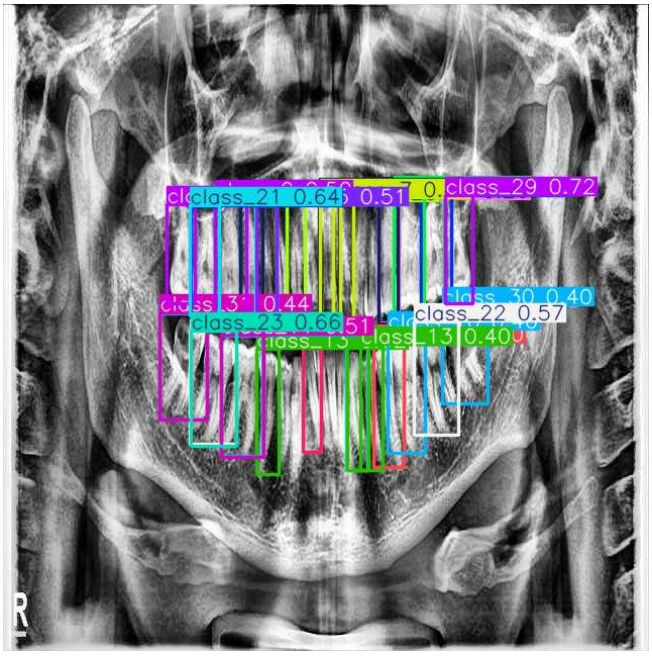
- High recall indicates the model successfully detects most teeth.
- Moderate precision suggests some misclassification between similar tooth classes.

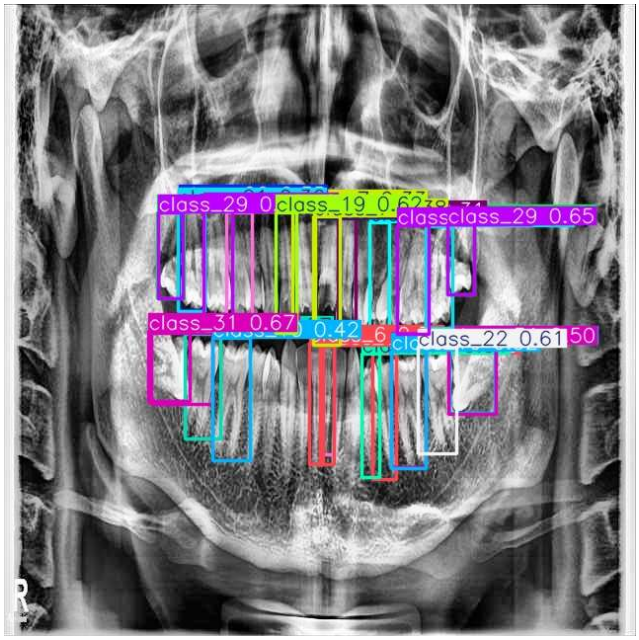
8. Observations

The model demonstrates good performance in detecting and localizing teeth across dental X-ray images, with high recall indicating that most teeth are successfully identified. However, precision is moderate, and some misclassifications are observed between adjacent or visually similar tooth classes. Additionally, overlapping bounding boxes and varying confidence scores suggest scope for improvement. Overall, the model shows stable learning and reasonable accuracy, with potential for enhancement through further training and data refinement.

9. Sample Outputs (Images)







The model successfully detects multiple teeth across different regions of the dental X-ray images. Bounding boxes are correctly localized in most cases, with moderate classification confidence scores. Some misclassifications are observed between adjacent tooth classes due to high visual similarity.